

Basic Linux Commands - Detailed Notes

1. man

The man command displays the manual page for a specific command. It provides information on usage, options, and examples.

- Example: man ls (Shows the manual for ls command)

2. ls, echo, read

- ls: Lists files and directories in the current directory.
 - Example: ls -l (Lists files with details)
- echo: Prints a message or variable to the terminal.
 - Example: echo "Hello, Linux!"
- read: Reads user input from the command line.
 - Example: read name; echo "Hello, \$name"

3. more, less, cat

- more: Displays file contents page by page.
 - Example: more filename
- less: Similar to more, but allows backward scrolling.
 - Example: less filename
- cat: Displays the entire file content.
 - Example: cat filename

4. cd, mkdir, pwd, find

- **cd:** Changes the directory.
 - Example: `cd /home/user`
- **mkdir:** Creates a new directory.
 - Example: `mkdir my_folder`
- **pwd:** Displays the current directory path.
 - Example: `pwd`
- **find:** Searches for files and directories.
 - Example: `find / -name "file.txt"`

5. mv, cp, rm, tar

- **mv:** Moves or renames files.
 - Example: `mv old.txt new.txt`
- **cp:** Copies files from one location to another.
 - Example: `cp file1.txt file2.txt`
- **rm:** Removes files or directories.
 - Example: `rm file.txt`
- **tar:** Archives multiple files into one.
 - Example: `tar -cvf archive.tar folder/`

6. wc, cut, paste

- **wc:** Counts words, lines, characters, or bytes.
 - Example: `wc file.txt`

- cut: Extracts specific columns from a file.
 - Example: `cut -d ' ' -f 1 file.txt`
- paste: Merges multiple files line by line.
 - Example: `paste file1.txt file2.txt`

7. head, tail, grep, expr

- head: Displays the first 10 lines of a file.
 - Example: `head file.txt`
- tail: Displays the last 10 lines of a file.
 - Example: `tail file.txt`
- grep: Searches for specific words or patterns in a file.
 - Example: `grep "error" logfile.txt`
- expr: Evaluates mathematical expressions.
 - Example: `expr 5 + 3`

8. chmod, chown

- chmod: Modifies file permissions.
 - Example: `chmod 755 script.sh`
- chown: Changes the file owner and group.
 - Example: `chown user:group file.txt`

9. Redirections & Piping

- `>` Redirects output to a file (overwrites if exists).

- Example: `ls > output.txt`
- `>>` Appends output to a file.
 - Example: `echo "Hello" >> file.txt`
- `|` Pipes output from one command to another.
 - Example: `ls | grep txt`

10. useradd, usermod, userdel, passwd

- `useradd`: Creates a new user account.
 - Example: `useradd newuser`
- `usermod`: Modifies an existing user account.
 - Example: `usermod -aG sudo newuser`
- `userdel`: Deletes a user account.
 - Example: `userdel newuser`
- `passwd`: Changes the password of a user.
 - Example: `passwd newuser`

11. df, top, ps

- `df`: Shows disk space usage.
 - Example: `df -h`
- `top`: Displays real-time system processes and resource usage.
 - Example: `top`

- `ps`: Lists currently running processes.
 - Example: `ps aux`

12. `ssh`, `scp`, `ssh-keygen`, `ssh-copy-id`

- `ssh`: Connects to a remote server securely.
 - Example: `ssh user@remote`
- `scp`: Securely copies files between systems.
 - Example: `scp file.txt user@remote:/path/`
- `ssh-keygen`: Generates SSH keys for secure authentication.
 - Example: `ssh-keygen -t rsa`
- `ssh-copy-id`: Copies SSH keys to a remote system for password-less login.
 - Example: `ssh-copy-id user@remote`